

Local Agentic RAG for PDFs

Cited answers from any PDF, in English, Hindi, Hinglish and Marathi, using open-weight models on a free Google Colab T4 GPU

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RESULTS AT A GLANCE (PROJECT NOTEBOOK RUN TOP TO BOTTOM, KAGGLE T4, 26 OCT)

76.9%	94.2%	88.9%	8.9 GB
answer accuracy	**page recall@5**	**multilingual accuracy**	**GPU memory in use**
61 English questions on 3 PDFs, up from 50.0% at baseline	evidence page in the top 5, up from 73.1% for the starting code	20 questions: Hindi, Hinglish, Marathi, English, scanned pages	of 15 GB on the T4, with the LLM, embedder and reranker loaded

1 The problem and the goal

Long PDFs such as statutes, budget documents, statistical abstracts and research reports are hard to search. Hosted chatbots can answer questions about them, but they cost money, send the document to a third party, and often answer confidently when the PDF does not say. We build for people who cannot or should not upload such files: law students and small law firms working with statutes and contracts, government and NGO analysts in India with budget books, statistical abstracts and scanned Hindi or Marathi circulars, researchers, and compliance teams. Our goal is a local tool that answers questions about *any* uploaded PDF, from a few pages to 10,000 pages, under four constraints:

- **Local only.** Open-weight models and no API keys, so the document never leaves the runtime.
- **Free hardware.** One Colab T4: 15 GB GPU memory, 12.7 GB host RAM, 2 vCPUs.
- **Grounded.** Every answer cites the pages it used, or says the document does not contain the answer.
- **Multilingual.** Questions and documents may be in different languages, and scanned pages must work.

We started from the reference notebook and an early codebase that had only a non-agentic baseline (MiniLM search, no tables, no OCR, English only). We rebuilt ingestion, retrieval and the agent, measured every change on a fixed benchmark, and then scaled ingestion to 10,000 pages and added a multi-user web app.

WORKING TODAY

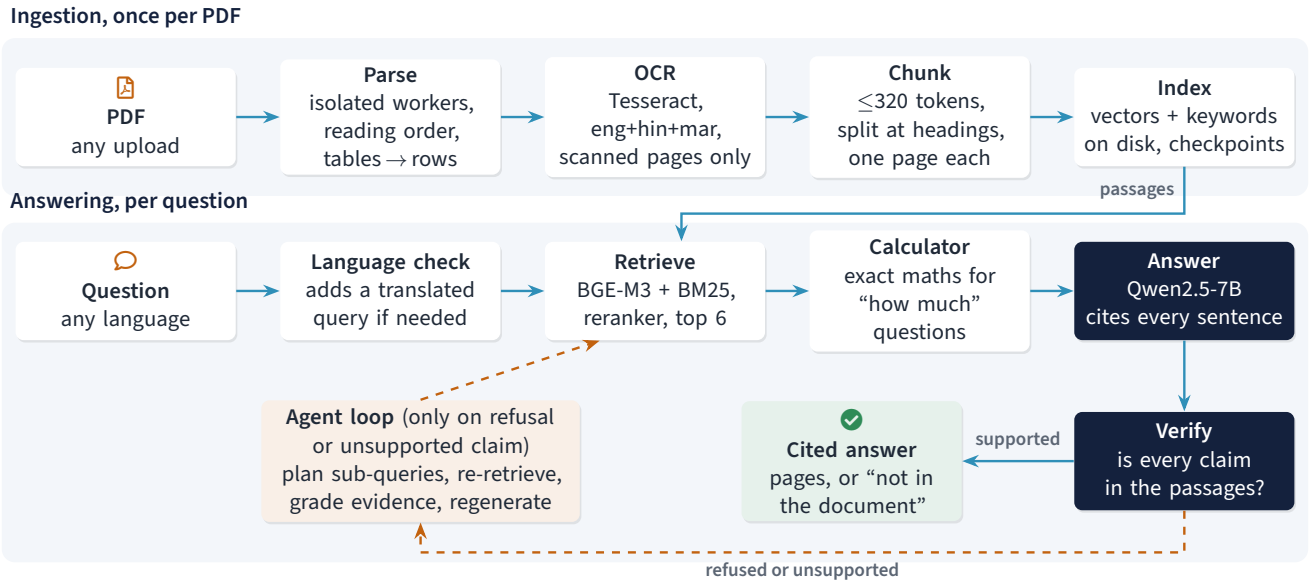
- Notebook that runs top to bottom on a fresh T4
- Web app: library, queue, citations that open the highlighted page
- PDFs of 100 to 10,000 pages, with crash recovery
- Tables, two-column layouts, scanned pages (OCR)

STILL OPEN

- Numerical questions on dense tables
- Five of 12 US Code questions still wrong (numeric, multi-hop, a definition)
- Faster ingestion of table-heavy PDFs

2 Approach and architecture

The system has two paths. **Ingestion** runs once per PDF and is cached by file hash. **Answering** runs per question. A single pass that retrieves, answers and verifies handles most questions, and the agent loop starts only when that pass refuses or the verifier finds an unsupported claim.



Navy boxes are LLM calls (Qwen2.5-7B-Instruct, 4-bit, served by Ollama). A typical question uses two to four calls, and the loop never exceeds eight. Structured steps use JSON-schema constrained output, and every step falls back to the simpler path if it fails.

Long documents and the web app

Ingestion is a two-pass stream. Pass 1 reads 16-page batches in parallel worker processes and records document-wide counts (repeated margin lines, body font size); pass 2 strips headers, chunks and embeds 48-page batches. A page that crashes or hangs the PDF library kills only its worker: it becomes an empty page with a warning. Both passes checkpoint, so a cancelled job or a closed notebook resumes where it stopped, and vectors (float16) and the BM25 matrix live on disk. The **web app** puts this behind a document library: uploads stream to disk, a background queue shows progress, and several users share one GPU with a visible waiting line. Answers stream word by word and are then replaced by the verified, cited text; a long document can be questioned once its first batch is indexed; each citation opens the rendered page with the passage highlighted; an optional access token and upload limits protect a shared server.

Design decisions and the evidence behind them

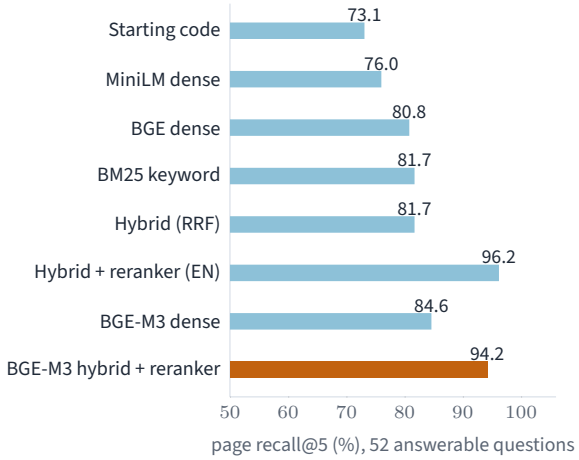
Decision	Why	Measured effect
Hybrid search + cross-encoder reranker	Dense search finds paraphrases, BM25 finds exact labels and numbers, and the reranker reads question and passage together.	Page recall@5 73.1% → 96.2% (English models)
Table rebuilding	Statistical tables are aligned with spaces, so plain text extraction scrambles rows and columns.	Rows read as `label | 2025 Q2 = 3.8`. Table questions rose from 42% to 83%, helped by better retrieval and table captions
BGE-M3 + multilingual reranker	English-only models miss Hindi and Hinglish wording.	Multilingual set: recall@5 0.94 → 1.00 and MRR 0.88 → 0.97, with answer accuracy tied
Adaptive agent, not an always-on loop	Our first always-on loop was *worse* than a single pass (23 vs 27 of 38): sub-queries crowded out the real evidence.	The loop runs only when needed, with fewer false refusals at similar accuracy
Python calculator	The 7B model subtracts and counts wrong even with the right numbers in view.	Arithmetic is exact and operands are checked against the cited row and column, but numerical questions are still 2/6
Tesseract CLI OCR at 300 dpi	The built-in OCR path lost about 30% of the Hindi words on scanned pages.	Hindi word recall 0.70 → 0.99 and 1.00

Engineering for a free runtime. The real limit was host RAM, not the GPU: the Ollama runner (the local LLM server, installed in 41 s on Colab; 96 s on Kaggle) grew by about 100 MB per request until Colab killed the session. Memory-mapped weights, one request slot, retries and a watchdog that restarts the runner fixed it. In the final run the watchdog restarted the runner and no question failed. Parsed documents are cached by file hash, so re-runs skip parsing.

3 Benchmarks

English set: 61 hand-written questions over three public PDFs: an ACL 2025 paper (17 pages, two columns), a U.S. BEA GDP release (28 pages, mostly statistical tables) and NIST AI 600-1 (64 pages). The questions cover seven categories, including 9 that the documents cannot answer. A correct answer must contain every required fact and must not refuse. **Multilingual set:** 20 questions over a Hindi article, a Marathi article, a scanned (image-only) Hindi document, and the English paper asked in Hindi and Hinglish. Final results: the project notebook run top to bottom on one T4 (Kaggle, 26 Oct, as Colab's free GPU quota was used up).

Retrieval: is the evidence page in the top 5?



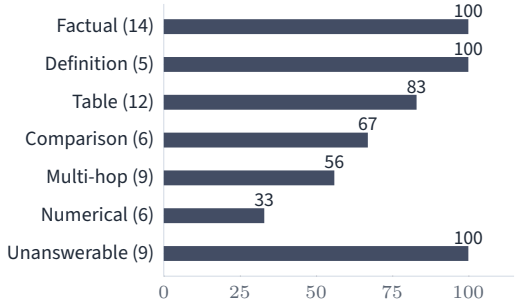
Orange is the default (multilingual). The reranker is the largest single gain. The multilingual default is one question below the English-only reranker here and ranks best at top-1 (68.3%).

Answers: end to end (Kaggle T4, 26 Oct)

	Baseline	Single pass	Agent
Answer accuracy	50.0%	71.2%	**76.9%**
Fact recall	53.2%	74.7%	80.1%
False refusals	26.9%	3.8%	1.9%
Unanswerable refused	9/9	9/9	9/9
Citation precision	70.6%	73.7%	78.4%
Latency, median	2.7 s	3.2 s	6.6 s

Same 61 questions and LLM in every column. "Baseline" is the starting design (MiniLM search, single pass), measured in an earlier T4 run. "Single pass" and "Agent" use the multilingual retriever. "Agent" is the default (adaptive loop + calculator). 95% intervals are about ±12 points, so small differences between the last two columns are within noise.

Agent accuracy by question type



Multilingual and scanned pages

Question slice	n	Correct
Hindi questions (8 answerable, 2 scanned)	9	7/8
Hinglish questions	3	3/3
Marathi questions (3 answerable)	4	3/3
English questions about Hindi/Marathi docs	4	3/4
Unanswerable, correctly refused	2	2/2
All answerable	18	**16/18 (88.9%)**

Answers come back in the question's language. The previous run of the same code scored 17/18, with both scanned-page (OCR) questions correct. English-only models also score 16/18 but retrieve worse (recall@5 0.94 vs 1.00).

Resources on a single T4 (Kaggle, 26 Oct)

GPU memory in use, all three models loaded	8.9 / 15.0 GB
Ollama install + 4.7 GB model download	3.8 min
Ingest a 17-page paper (first time, cached after)	8.5 s
Scanned 2-page Hindi document, with OCR	16 s
Retrieval ablation, 7 configurations	1.9 min
English answers, 2 configurations × 61	14.3 min
Multilingual answers, 2 configurations × 20	7.4 min
Offline test suite (current code)	845 tests

What an answer looks like


Q: इस शोध में लेखकों ने किस डेटासेट का पुनः उपयोग किया?

Verified: every claim supported Pages 2, 3

इस शोध में लेखकों ने Re-DocRED डेटासेट का पुनः उपयोग किया [3][4][2].

Sources

[2] p. 2 · arxiv-2503-05037
"In all cases, retrievers favor D1 over D2. tioning, we repurpose a document-level relation extraction dataset (Re-DocRED (Tan et al., 2022)). We first investigate biases individually, identifying tend…"

[3] p. 3 · arxiv-2503-05037
"To gain fine-grained control over the facts present in a document, we take a novel approach by repurposing a relation extraction dataset that provides Table 2: Models' performance on NQ dataset with t…"

[4] p. 2 · arxiv-2503-05037
"Table 1: Construction of query–document pairs. From each Re-DocRED article, we select an Evidence Sentence (Sev) that contains both the Head Entity (h, the relation's subject) and Tail Entity (t, its…"

Screenshot from the T4 run. A Hindi question about the English paper ("Which dataset did the authors repurpose?") is answered in Hindi. The answer names Re-DocRED, cites pages 2 and 3, and the verifier marks every claim as supported. The quoted sources are the passages the citations point to.

Long documents (100 to 10,000 pages)

A fictional statistical compendium was generated at six sizes (ruled tables on 58% of pages, two-column pages, footnotes, about 1.5% scanned pages) with 40 independently written questions per size, each fact planted at a known page. **This synthetic corpus shows how the system scales, not how accurate it is on real documents:** its prose is template text, and the same questions are used at every size so that sizes can be compared.

Pages	Ingest (GPU tables)	Ingest (CPU)	Recall@5	Accuracy	Answer p50	Index on disk
100	30 s	55 s	0.972	0.750	6.3 s	2 MB
250	58 s	93 s	0.972	0.722	6.3 s	5 MB
1,000	3.8 min	5.7 min	0.944	0.722	6.1 s	19 MB
2,500	9.3 min	14.7 min	0.944	0.694	6.8 s	47 MB
5,000	18.2 min	31.4 min	0.917	0.750	6.3 s	94 MB
10,000	36.9 min	71.3 min	0.889	0.722	6.0 s	189 MB

Kaggle T4, final code, 27 Oct (run 4), 2 parse workers (the free-Colab CPU count). CPU column: the 26 Oct run, with identical recall and accuracy at every size. Accuracy is on the 36 answerable questions per size. Peak RAM up to 9.4 GB; at every size a killed parse worker cost one page and an interrupted job resumed from its checkpoint.

Real documents (46 independent questions, Kaggle T4; 26 Oct runs v2–v4 and the final run, 27 Oct)

	Pages	Ingest	Answerable correct	Unanswerable refused
9/11 Report	585	46 s	11/12 → **12/12**	1/3
Stat. Abstract	940	15.7 min	9/13 → **11/13**	3/3
US Code T26	4,211	11.7 min	6/12 → **7/12**	2/3
Total			**30/37** (was 26)	6/9

Before → final run (27 Oct; GPU table detection, one document at a time). Earlier ingest (26 Oct): 61 s, 23.8 min, 15.5 min, with CPU table detection and two documents at once, so contention is included. Title 26's 7/12 equals run 2 (the section fix was designed on this set, so not held out). Budget Appendix (no questions): 24.0 min (33.9 before).

Recovery. At every size, a parse worker was killed on one page and the job was cancelled half-way, then restarted (Kaggle T4, 26 Oct). Each time the crash cost exactly that one page, the job resumed from its checkpoint, and the index matched a clean run. At 10,000 pages the interrupted job took 4,338 s against 4,279 s clean (+1.4%).

Several users. The real web server, two answers at a time, on the 1,000-page document with 1, 2, 4 and 8 users (Kaggle T4, 26 Oct, 3 questions per user): 0 failed requests of 45. Throughput saturates at about 10.5 answers per minute. Median latency rises from 12.8 s (1 user) to 44.6 s (8 users) as users wait in a visible queue.

4 What was delivered

BUILT AND MEASURED ON THE GPU (KAGGLE T4, 26 OCT)

- Ingestion of 100 to 10,000 pages: two-pass stream, checkpoints, crash-isolated workers, on-disk index
- Web app: library, background queue, citations that open the highlighted page; 0 failures with 1 to 8 users
- Hybrid retrieval with reranking; adaptive agent with verifier and calculator
- Hindi, Hinglish, Marathi and scanned pages (OCR)
- Page references in questions ("page 4,512"): such pages now rank 3 to 4 instead of 87 to 895 before reranking
- Independent evaluation: 240 synthetic and 46 real-document questions written by a separate author
- Notebook runs top to bottom in 32.7 min with no errors
- Access token and upload limits; 845 offline tests

NEW THIS WEEK: MEASURED ON THE GPU

Feature	What exists	Measured on a T4
Token streaming	Ollama streaming, batched server events; demo server sends queued, started, steps, tokens, answer	First words after 0.47 s median with 1 user, 0.70 s with 2 (T4, 26 Oct)
Ask while processing	Askable after the first indexed batch; final index byte-identical with or without partial queries (test)	1,000-page PDF askable 206 s after upload (96 pages indexed); answer correct and cited (T4, 26 Oct)
"Which table" naming	Appends the cited table's printed number when the answer omits it; no prompt change	Statistical Abstract "which table" questions 5/6 correct (T4, 27 Oct)
Layout model on the GPU	Opt-in setting for table-heavy PDFs (about 80% of ingestion time)	1,000-page PDF: 119.0 s, not 224.7 s (1.89×), same output. Stat. Abstract parse 8.7 min (17.5 min on CPU, two documents at once)
Final re-run of the 46 real questions	Run once more at the end, after the fixes above	30/37 answerable correct (26/37 before), 6/9 unanswerable refused (T4, 27 Oct)

Docker image built; install smoke test passed with the real model on a CPU host (27 Oct); GPU in Docker not yet tested. CI passes.

Latest improvements: before and after

We analysed every failed question, fixed only what the evidence pointed to, added regression tests for each fix, and measured every change on the GPU before keeping it.

Change	Evidence that it was needed	Before → after
OCR through the Tesseract CLI	One of the two scanned-page questions failed because the OCR text had lost words	Word recall 0.70 → 0.99 and 1.00. Scanned-page questions 1/2 → 2/2
Refusal detection	An unanswerable question got a cited "the document does not say…" that was scored as an answer	Unanswerable refused 8/9 → 9/9
Calculator grounding	It took operands from the wrong table row and overrode a stated "+$6.8 billion"	English 73.1% → 75.0%. Numerical questions still 2/6
Cited answers ending in "not found"	A correct, cited Hindi answer followed by the marker was scored as a refusal	Multilingual 83.3% → 94.4% in that run
Table captions and section labels	Look-alike arXiv tables could not be told apart. BEA rows carried the previous table's title	English 75.0% → 76.9%. Table questions 9/12 → 10/12. One BEA question now ranks lower

One change made results worse and was reverted: a stricter "refuse unless the exact fact is present" prompt rule raised false refusals from 3.8% to 21.2% in a single-pass run. Net effect of this week's changes on the English set: 73.1% → 76.9% (four questions fixed, two regressed).

5 Limitations

- **Ingestion of table-heavy PDFs is slow on 2 CPUs:** 2.3 to 2.9 pages/s, so 10,000 pages take about 71 min and the 1,224-page Budget Appendix 34 min (37 min for 10,000 pages with the opt-in GPU table detection). The table model is about 80% of that. A faster line-only detector invented tables in academic papers, so it stays off.
- **Section questions on long legal codes.** On the US Code (Title 26) a bold section heading was directly followed by a bold sub-heading and chunks kept only the latest, so only 355 sections were identified. Each chunk now records its section number: 1,877 sections, page recall@5 0.75 → 0.92, answerable 6/12 → 7/12 (run 2; not held out, the fix was designed on this set). §164, §4081, §6501 and §7203 are now correct; 5 of 12 are still wrong (numeric, multi-hop, a definition). On a new legal code (US Code Title 21; held-out questions, single use, 27 Oct) 19/25 answerable are correct, 4/5 unanswerable refused, page recall@5 0.917, and 8 of the 9 questions naming a section are correct.
- **Questions that presuppose something absent** (an act passed after the report) are sometimes answered from the nearest passage: 6 of 9 unanswerable real-document questions were refused. A stricter refusal rule raised false refusals from 3.8% to 21.2%, so it was not kept.
- **Small benchmarks.** 61 English and 20 multilingual questions from one annotator; 46 real-document questions; grading matches required facts and can miss a paraphrase. The synthetic corpus is template prose, and the real documents are four US federal PDFs.
- **Numerical questions** remain the weakest category (2/6): the 7B model often picks a number from the wrong row or column of a dense table.
- **Coverage.** Hindi, Hinglish and Marathi were measured on 20 questions; OCR on one 2-page scan; charts and figures are not interpreted.
- **Shared use.** The token has no rate limit on guesses, and users behind a proxy share one upload slot. The phone layout was checked at 390 px in an emulated phone browser, not on a real phone.

6 Next steps

After the hackathon: cell-level table retrieval and a larger model (Qwen2.5-14B fits the measured headroom) for number extraction only; check the section fix on more legal codes; a larger multi-annotator benchmark with an LLM judge, including Hindi and Marathi long documents; rate limiting before any use beyond one trusted group.